# A Technical Introduction to Foundation model AI

## Section 1

The foundation model developer itself will then take the data and use the supplied compute to actually train the foundation model. After the foundation model is completely built, much of the data and labor requirements abate. In this development process, hardware and compute are the most necessary, and also the most exclusive resources. To train larger and more complex AI, a sufficient amount of compute is key. However, compute is consolidated in the hands of a few, select entities, which most foundation model developers depend on. As such, the foundation model pipeline is concentrated heavily around these providers. Compute is also costly; in 2023, AI companies spent more than 80% of total capital on compute resources. Foundation models require a large amount of general data to power their capabilities. Early foundation models scraped from subsets of the internet to provide this data information. As the size and scope of foundation models grows, larger quantities of internet scraping becomes necessary, resulting in higher likelihoods of biased or toxic data. This toxic or biased data can disproportionately harm marginalized groups and exacerbate existing prejudices. To address this issue of low-quality data that arose with unsupervised training, some foundation model developers have turned to manual filtering. This practice, known as data labor, comes with its own host of issues. Such manual data detoxification is often outsourced to reduce labor costs, with some workers making less than $2 per hour. The foundation model will then be hosted online either via the developer or via an external organization. Once released, other parties can create applications based on the foundation model, whether through fine-tuning or wholly new purposes. People can then access these applications to serve their various means, allowing one foundation model to power and reach a wide audience.

## Section 2

History Quanta Magazine traced world models back to a 1943 publication by Kenneth Craik on mental models and the blocks world of SHRDLU in the 1960s. Business Insider traced world models to a 1971 paper by Jay Wright Forrester. A related idea of organizing world knowledge, the frame representation, was proposed by Marvin Minsky in 1974. In 2018, researchers David Ha and Jürgen Schmidhuber defined world models in the context of reinforcement learning: an agent with a variational autoencoder model V for representing visual observations, a recurrent neural network model M for representing memory, and a linear model C for making decisions. They suggested that agents trained on world models in environments that simulate reality could be applied to real world settings. In 2022, Yann LeCun saw a world model (defined by him as a neural network that acts as a mental model for aspects of the world that are seen as relevant) as part of a larger system of cognitive architecture – other neural networks that are analogous to different regions of the brain. In his view, this framework could lead to commonsense reasoning. LeCun has estimated that world models would be fully functional by the late 2020s to mid 2030s.

## Section 3

Systems The size of foundation models also brings about issues with the computer systems they run on. The average foundation model is too large to be run within a single accelerator's memory and the initial training process requires an expensive amount of resources. Such issues are predicted to further exacerbate in future as foundation models grow to new heights. Due to this constraint, researchers have begun looking into compressing model size through tight model inference. GPUs are the most common choice of compute hardware for machine learning, due to high memory storage and strong power. Typical foundation model training requires many GPUs, all connected in parallel with fast interconnects. Acquiring a sufficient amount of GPUs of requisite compute efficiency is a challenge for many foundation model developers, one that has led to an increasing dilemma in the field. Larger models require greater compute power, but often at the cost of improved compute efficiency. Since training remains time-consuming and expensive, the tradeoff between compute power and compute efficiency has led only a few select companies to afford the production costs for large, state of the art foundation models. Some techniques like compression and distillation can make inference more affordable, but they fail to completely shore up this weakness.

## Section 4

Definitions The Stanford Institute for Human-Centered Artificial Intelligence's (HAI) Center for Research on Foundation Models (CRFM) coined the term "foundation model" in August 2021 to mean "any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks". This was based on their observation that preexisting terms, while overlapping, were not adequate, stating that "'large language model' was too narrow given the focus is not only language; 'self-supervised model' was too specific to the training objective; and 'pretrained model' suggested that the noteworthy action all happened after 'pretraining." The term "foundation model" was chosen over "foundational model" because "foundational" implies that these models provide fundamental principles in a way that "foundation" does not. The term vision-language model (VLM) is also used as a near-synonym. As governments regulate foundation models, new legal definitions have emerged.

## Section 5

Concerns TechCrunch noted that world models could use more data than large language models and would require significantly more computational power (including the use of thousands of GPUs for training and inference). It also noted the risk of hallucinations, coverage bias and algorithmic bias. Similarly, The Financial Times noted the difficulty and expense in collecting data to simulate the world and training models to use that data. Creative professionals have expressed concern that world models could disrupt jobs in their industries. Other concerns include data privacy, simulation of harmful situations, misinformation and disinformation, emergent behaviors, and copyright.

## Section 6

In artificial intelligence, a foundation model (FM), also known as large x model (LxM, where "x" is a variable representing any text, image, sound, etc.), is a machine learning or deep learning model trained on vast datasets so that it can be applied across a wide range of use cases. Generative AI applications like large language models (LLM) are common examples of foundation models. Building foundation models is often highly resource-intensive, with the most advanced models costing hundreds of millions of dollars to cover the expenses of acquiring, curating, and processing massive datasets, as well as the compute power required for training. These costs stem from the need for sophisticated infrastructure, extended training times, and advanced hardware, such as GPUs. In contrast, adapting an existing foundation model for a specific task or using it directly is far less costly, as it leverages pre-trained capabilities and typically requires only fine-tuning on smaller, task-specific datasets. Early examples of foundation models are language models like OpenAI's GPT series and Google's BERT. Beyond text, foundation models have been developed across a range of modalities—including DALL-E, Stable diffusion, and Flamingo for images, MusicGen and LLark for music, and RT-2 for robotic control. Foundation models are also being developed for fields like astronomy, radiology, ophthalmology, genomics, coding, time series forecasting, mathematics, and chemistry.